Abstract:

In this Project, I will setup the process for Selenium automation testing, a powerful tool for automating web browser interactions. To start, I first installed Python3 which is essential for running Selenium scripts. Once Python was successfully installed, I proceeded to install the Selenium framework by executing the command `**pip install selenium**` in the command prompt. This ensures that we have the Selenium libraries required for our testing.Next, I installed the **Chrome WebDriver**, a critical component that allows Selenium to interact with the Chrome browser. After downloading and placing the WebDriver in a designated directory, I set up the environment path by editing the system variables. This step involved adding the WebDriver's folder location to the system's path, making it accessible for Selenium. I have also installed some library for our testing, for example, **Pymysql, venv** etc

I have tested some features of my Web Technologies project. The features are **Login, Signup, Employee Add, Merchant Add** by showing the possible outcomes from testing.I have also shown the changes in Database where applicable.

Python Install:



Fig_1: Python Install

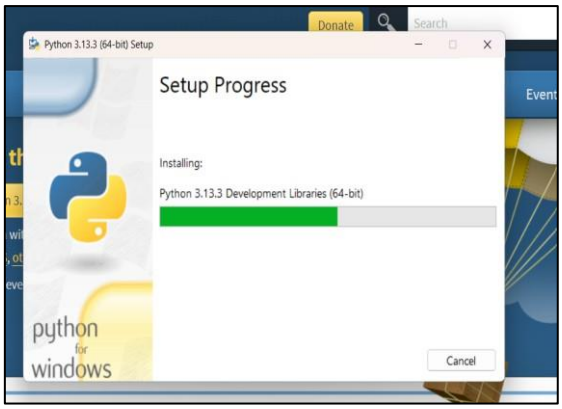


Fig_2: Python Install

Installation check:

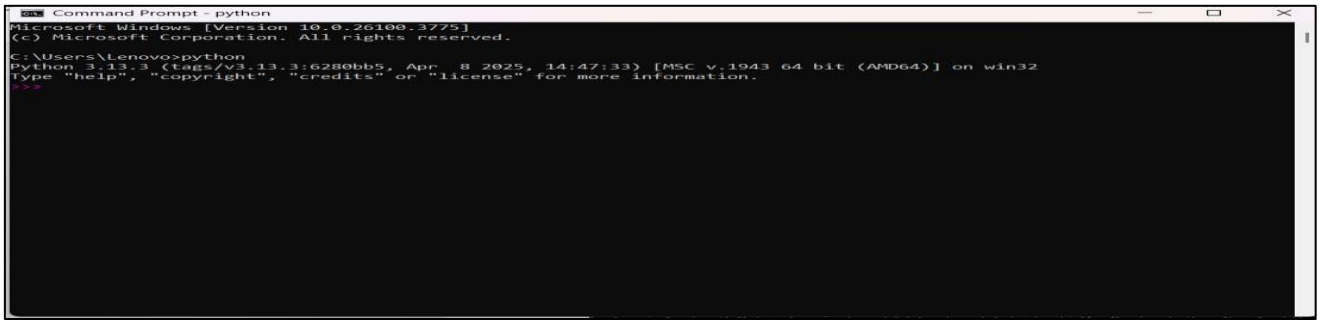


Fig_3: Installation Check

Install selenium framework in python:

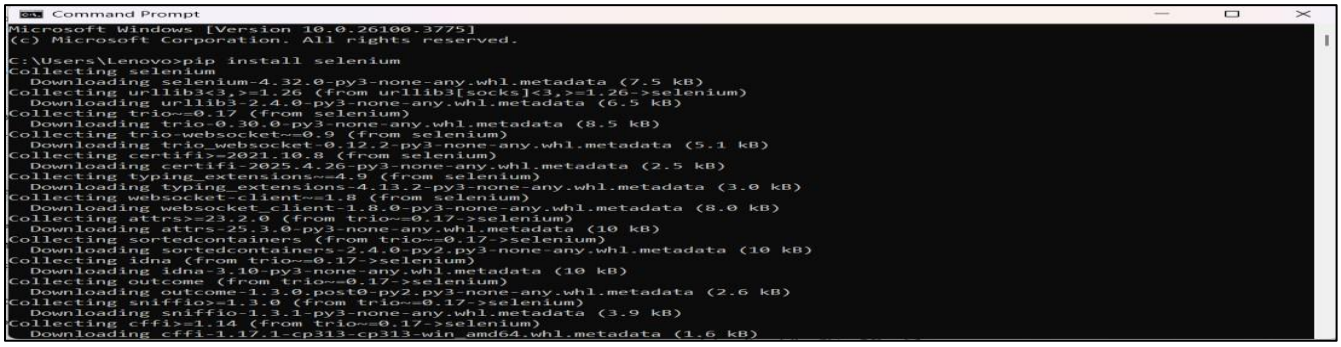


Fig_4: Install selenium framework in python

Chrome WebDriver install:

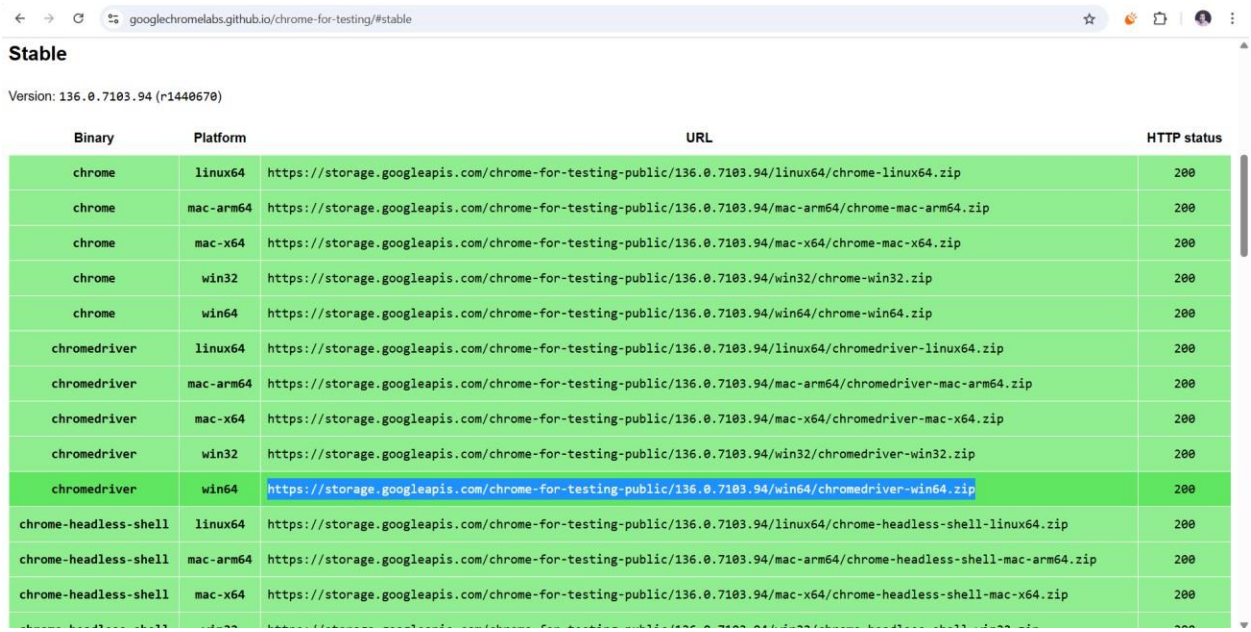


Fig_5: Chrome WebDriver install

Chrome WebDriver environment Path setup:

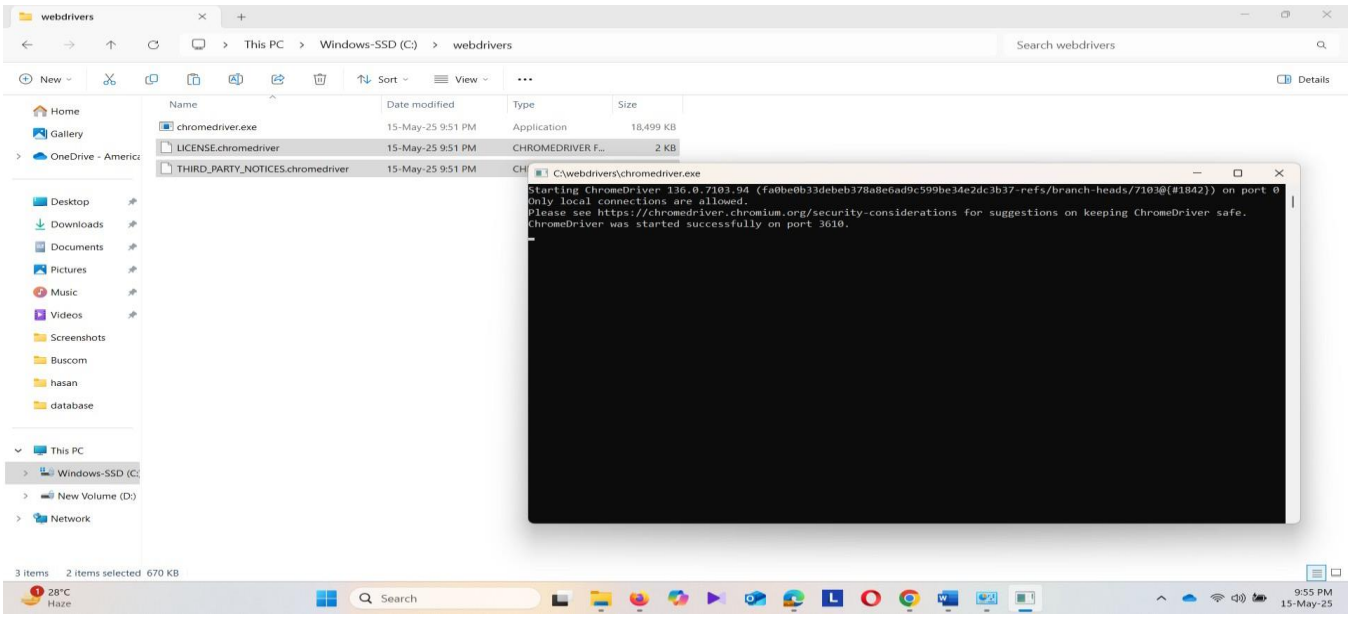


Fig_6: Chrome WebDriver environment Path setup

Extensions for python in VS CODE:

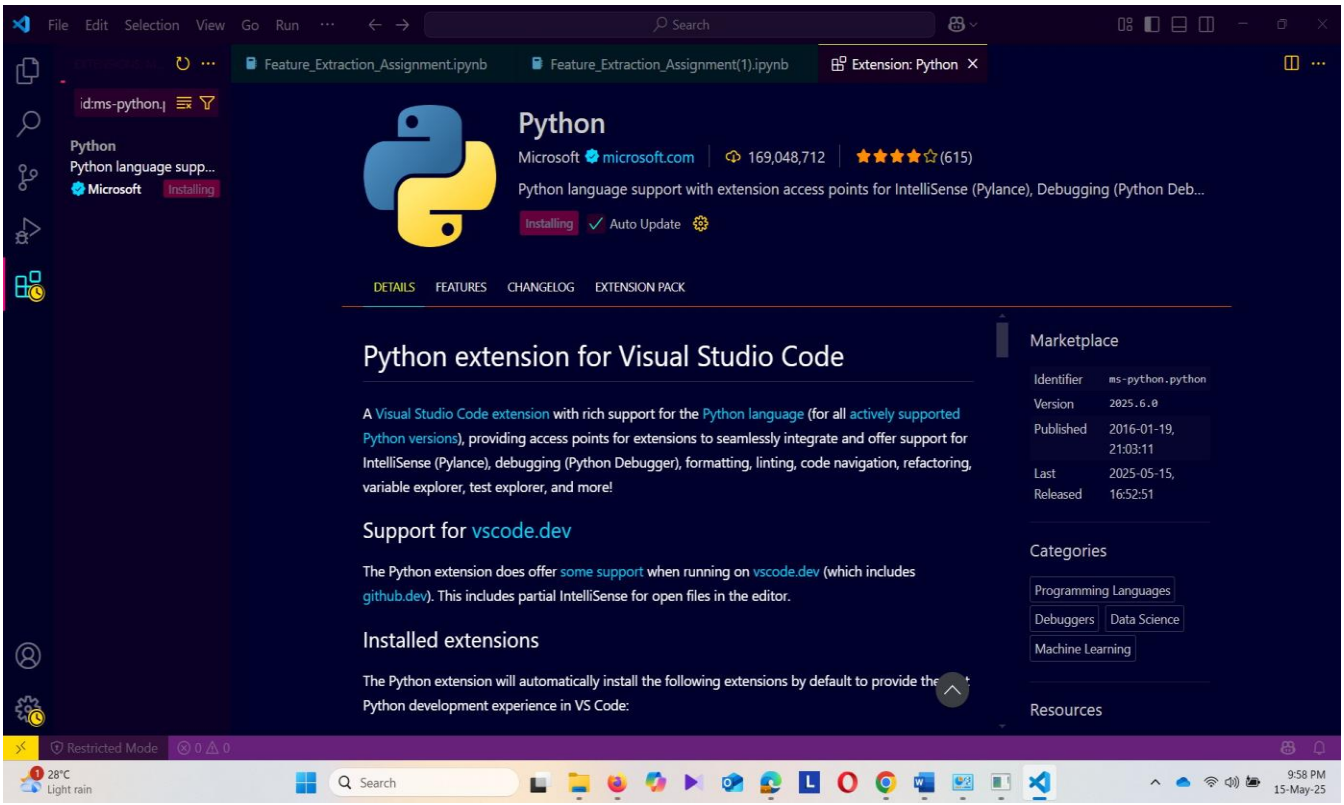
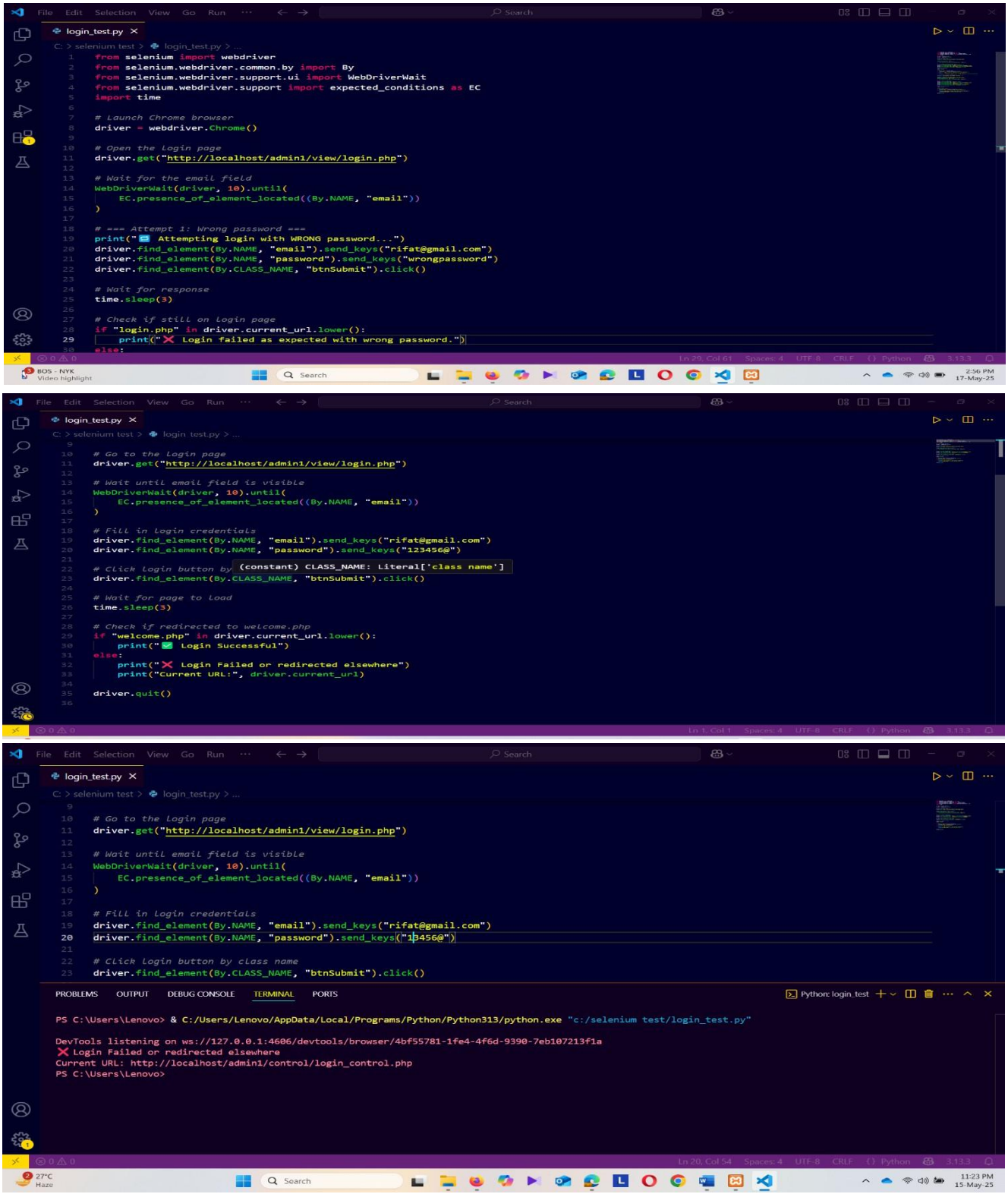


Fig_7: Extensions for Python in VS Code

Test Case 1: Login check:

Selenium script:



```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time

# Launch Chrome browser
driver = webdriver.Chrome()

# Open the login page
driver.get("http://localhost/admin1/view/login.php")

# Wait for the email field
WebDriverWait(driver, 10).until(
    EC.presence_of_element_located((By.NAME, "email"))
)

# === Attempt 1: Wrong password ===
print("🔄 Attempting login with WRONG password...")
driver.find_element(By.NAME, "email").send_keys("rifat@gmail.com")
driver.find_element(By.NAME, "password").send_keys("wrongpassword")
driver.find_element(By.CLASS_NAME, "btnSubmit").click()

# Wait for response
time.sleep(3)

# Check if still on login page
if "login.php" in driver.current_url.lower():
    print("❌ Login failed as expected with wrong password.")
else:
```

```
# Go to the Login page
driver.get("http://localhost/admin1/view/login.php")

# Wait until email field is visible
WebDriverWait(driver, 10).until(
    EC.presence_of_element_located((By.NAME, "email"))
)

# Fill in login credentials
driver.find_element(By.NAME, "email").send_keys("rifat@gmail.com")
driver.find_element(By.NAME, "password").send_keys("123456@")

# Click login button by (constant) CLASS_NAME: Literal['class name']
driver.find_element(By.CLASS_NAME, "btnSubmit").click()

# Wait for page to load
time.sleep(3)

# Check if redirected to welcome.php
if "welcome.php" in driver.current_url.lower():
    print("✅ Login Successful")
else:
    print("❌ Login Failed or redirected elsewhere")
    print("Current URL:", driver.current_url)

driver.quit()
```

```
# Go to the Login page
driver.get("http://localhost/admin1/view/login.php")

# Wait until email field is visible
WebDriverWait(driver, 10).until(
    EC.presence_of_element_located((By.NAME, "email"))
)

# Fill in login credentials
driver.find_element(By.NAME, "email").send_keys("rifat@gmail.com")
driver.find_element(By.NAME, "password").send_keys("13456@")

# Click login button by class name
driver.find_element(By.CLASS_NAME, "btnSubmit").click()
```

```
PS C:\Users\Lenovo> & C:/Users/Lenovo/AppData/Local/Programs/Python/Python313/python.exe "c:/selenium test/login_test.py"

DevTools listening on ws://127.0.0.1:4606/devtools/browser/4bf55781-1fe4-4f6d-9390-7eb107213f1a
❌ Login Failed or redirected elsewhere
Current URL: http://localhost/admin1/control/login_control.php
PS C:\Users\Lenovo>
```

Fig_8: Terminal View of Failed Login

Successful login goes to dashboard (Terminal view):

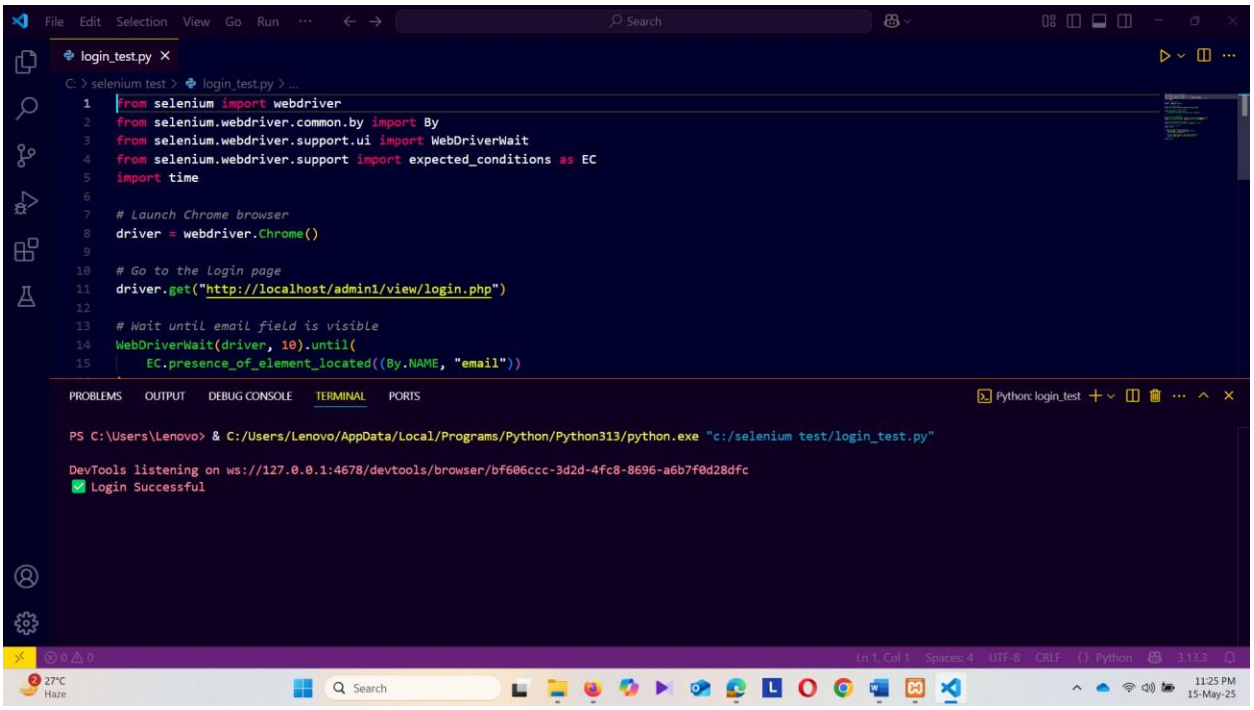


Fig_9: Terminal View of Successful Login (Dashboard Redirect)

The web Page:

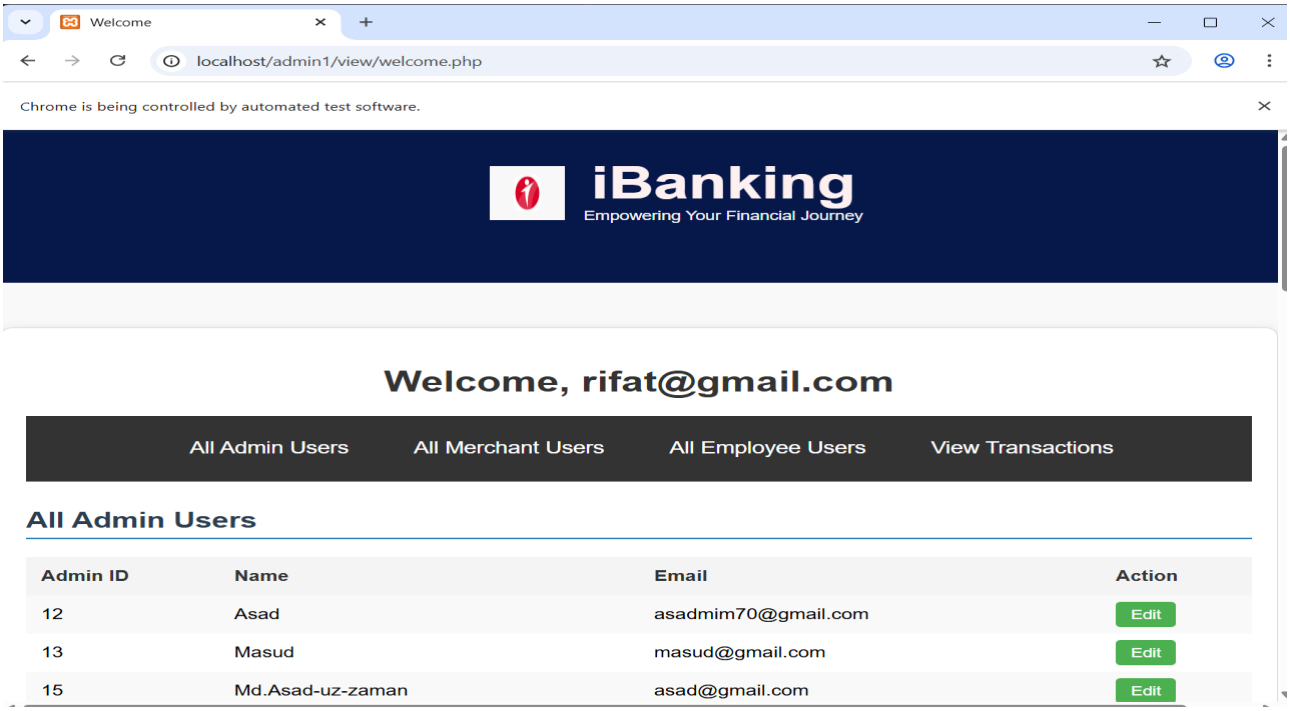
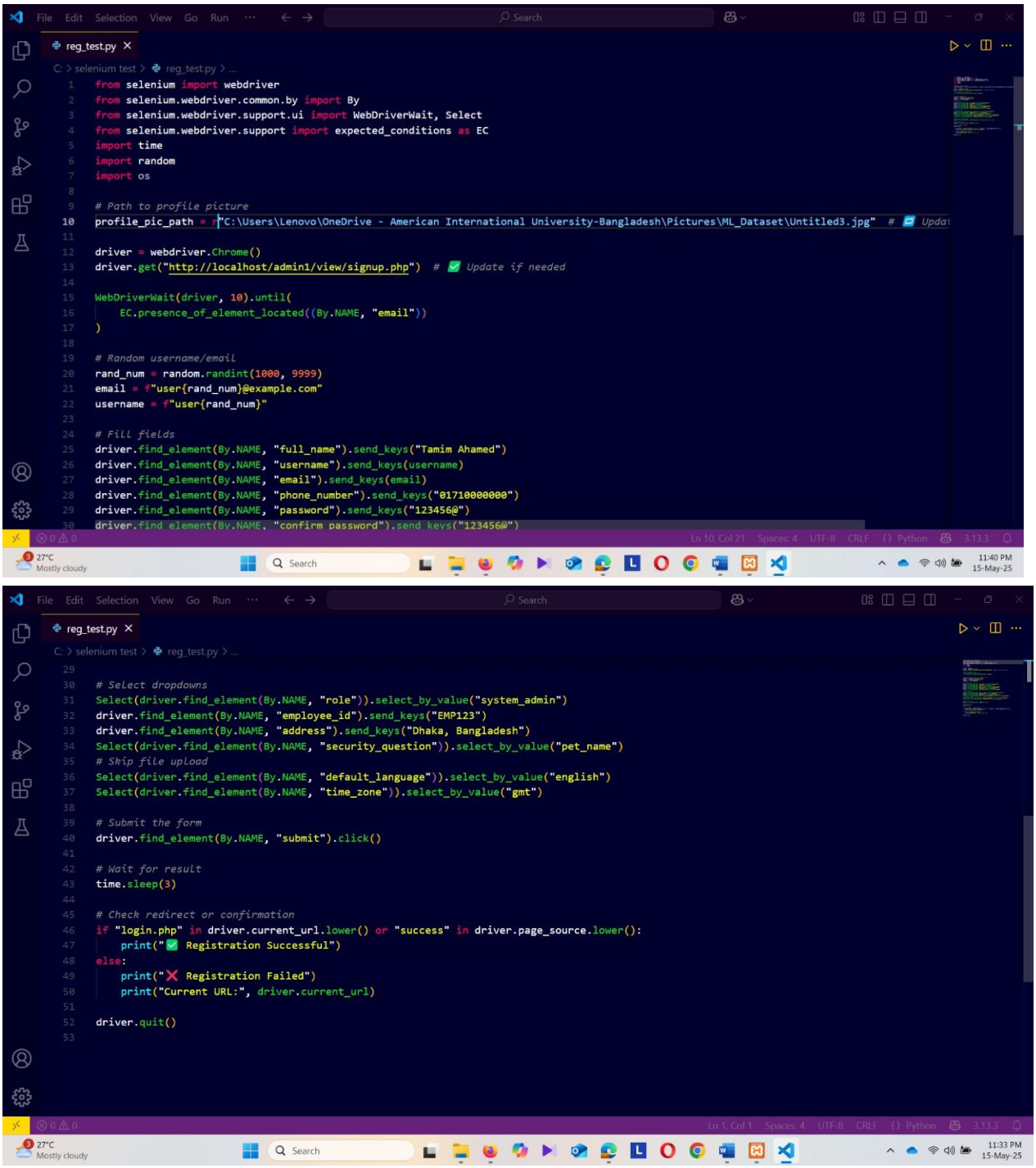


Fig_10: Login Web Page

Test Case 2: Registration check [selenium script]



```python
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait, Select
from selenium.webdriver.support import expected_conditions as EC
import time
import random
import os

# Path to profile picture
profile_pic_path = r"C:\Users\Lenovo\OneDrive - American International University-Bangladesh\Pictures\ML_Dataset\Untitled3.jpg"  # Updat

driver = webdriver.Chrome()
driver.get("http://localhost/admin1/view/signup.php")  # ✅ Update if needed

WebDriverWait(driver, 10).until(
    EC.presence_of_element_located((By.NAME, "email"))
)

# Random username/email
rand_num = random.randint(1000, 9999)
email = f"user{rand_num}@example.com"
username = f"user{rand_num}"

# Fill fields
driver.find_element(By.NAME, "full_name").send_keys("Tamim Ahamed")
driver.find_element(By.NAME, "username").send_keys(username)
driver.find_element(By.NAME, "email").send_keys(email)
driver.find_element(By.NAME, "phone_number").send_keys("01710000000")
driver.find_element(By.NAME, "password").send_keys("123456@")
driver.find_element(By.NAME, "confirm_password").send_keys("123456@")
```

```python
# Select dropdowns
Select(driver.find_element(By.NAME, "role")).select_by_value("system_admin")
driver.find_element(By.NAME, "employee_id").send_keys("EMP123")
driver.find_element(By.NAME, "address").send_keys("Dhaka, Bangladesh")
Select(driver.find_element(By.NAME, "security_question")).select_by_value("pet_name")
# Skip file upload
Select(driver.find_element(By.NAME, "default_language")).select_by_value("english")
Select(driver.find_element(By.NAME, "time_zone")).select_by_value("gmt")

# Submit the form
driver.find_element(By.NAME, "submit").click()

# Wait for result
time.sleep(3)

# Check redirect or confirmation
if "login.php" in driver.current_url.lower() or "success" in driver.page_source.lower():
    print("✅ Registration Successful")
else:
    print("❌ Registration Failed")
    print("Current URL:", driver.current_url)

driver.quit()
```

Fig_11: Registration Check

Registration page (web view):

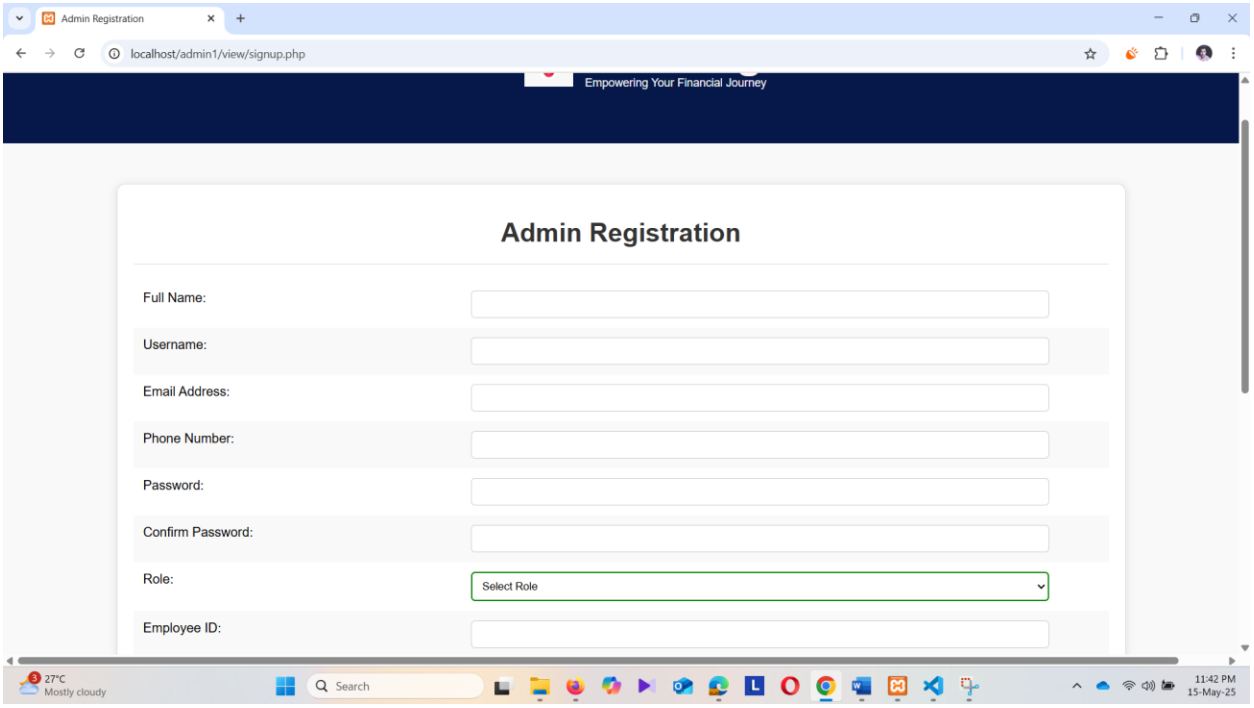
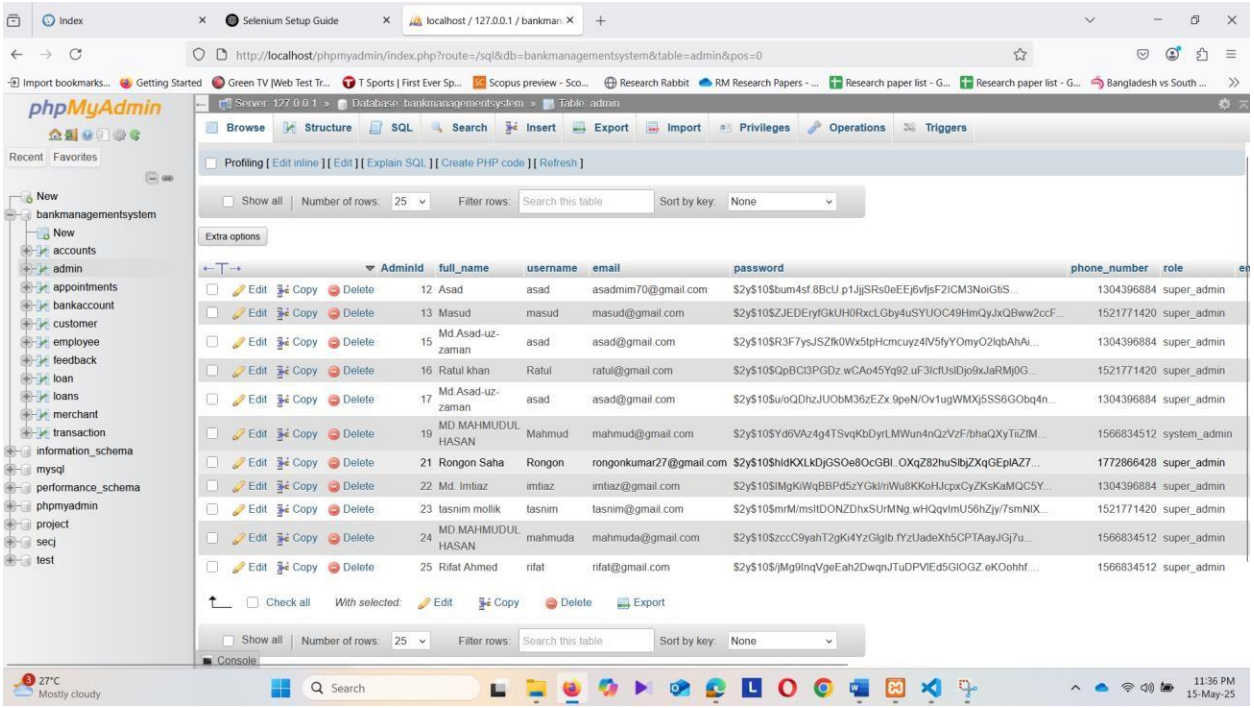


Fig_12: Registration Page (Web View)

Before registering new user database:



Upon Successful registration; user[tamim] gets inserted to database:

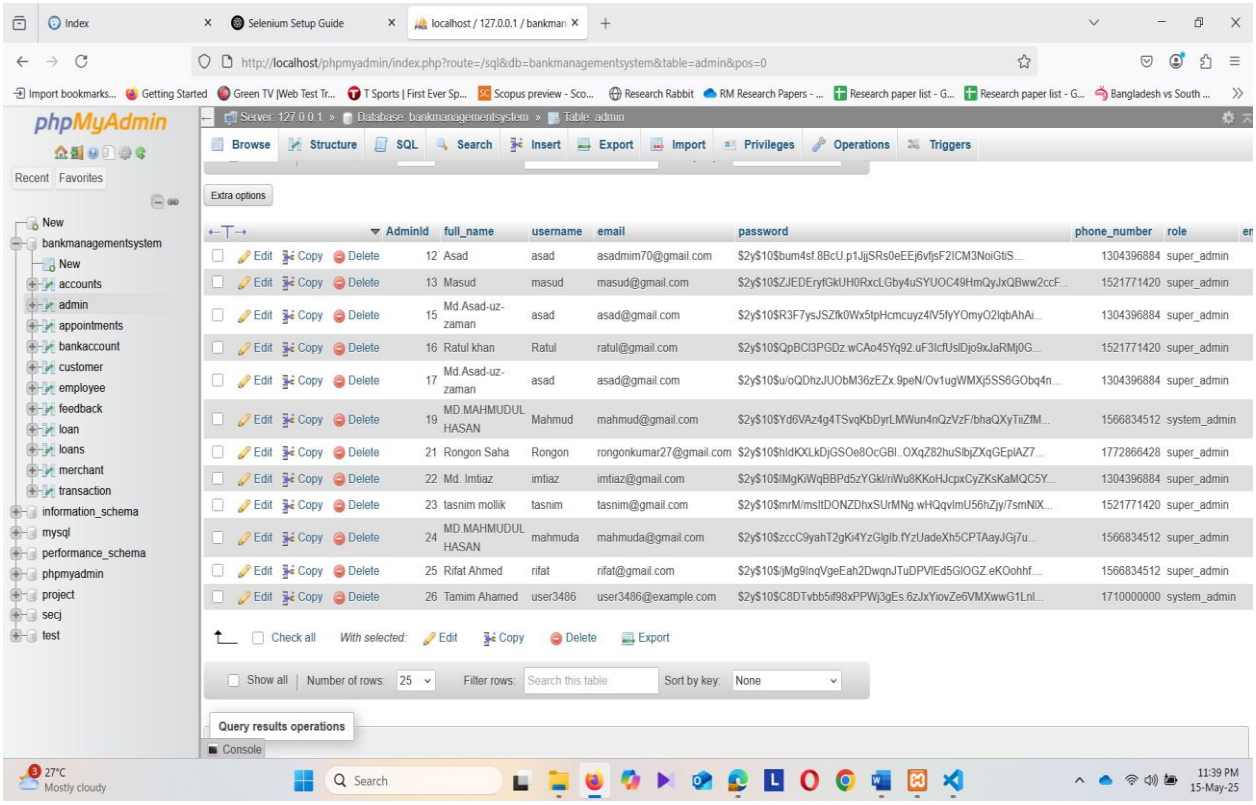


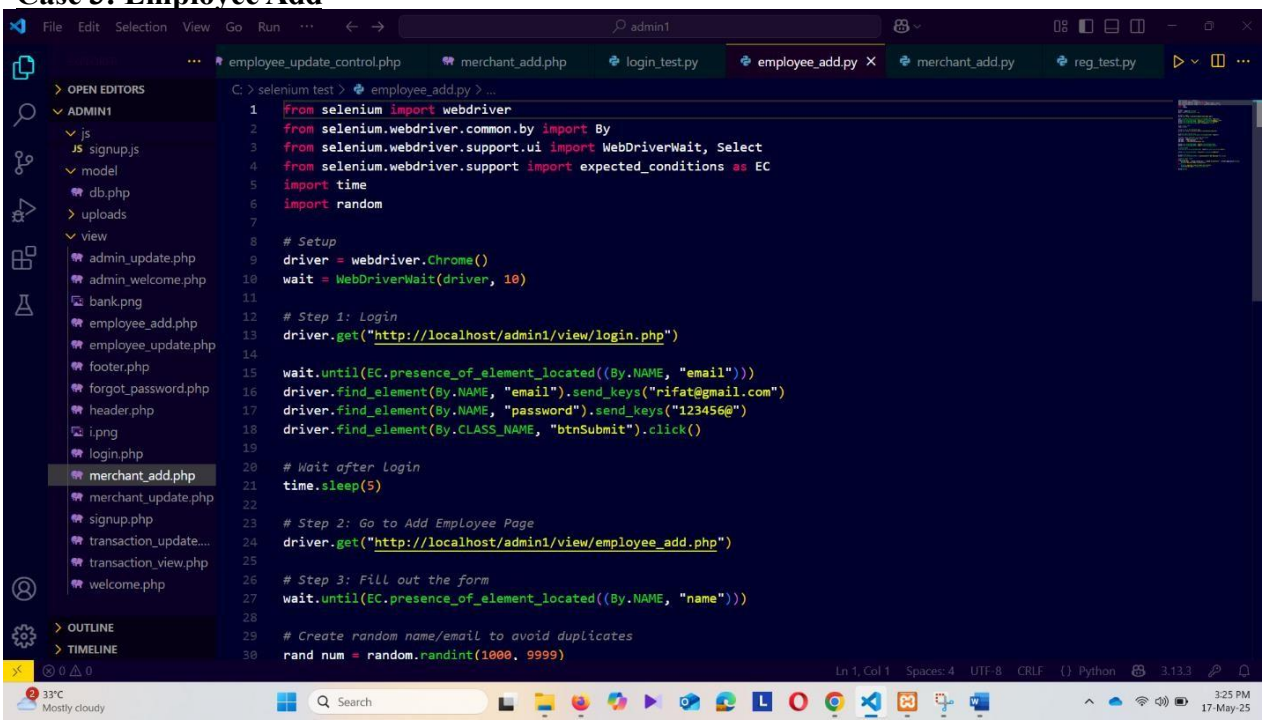
Fig_13: Database View After Successful Registration

Terminal view:

```
DevTools listening on ws://127.0.0.1:4888/devtools/browser/79397c8f-a17d-4cbc-8ece-282857e3a3ef
✅ Registration Successful (with file upload)
PS C:\Users\Lenovo>
```

Fig_14: Registration Terminal Output

Case 3: Employee Add



```python
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait, Select
from selenium.webdriver.support import expected_conditions as EC
import time
import random

# Setup
driver = webdriver.Chrome()
wait = WebDriverWait(driver, 10)

# Step 1: Login
driver.get("http://localhost/admin1/view/login.php")

wait.until(EC.presence_of_element_located((By.NAME, "email")))
driver.find_element(By.NAME, "email").send_keys("rifat@gmail.com")
driver.find_element(By.NAME, "password").send_keys("123456@")
driver.find_element(By.CLASS_NAME, "btnSubmit").click()

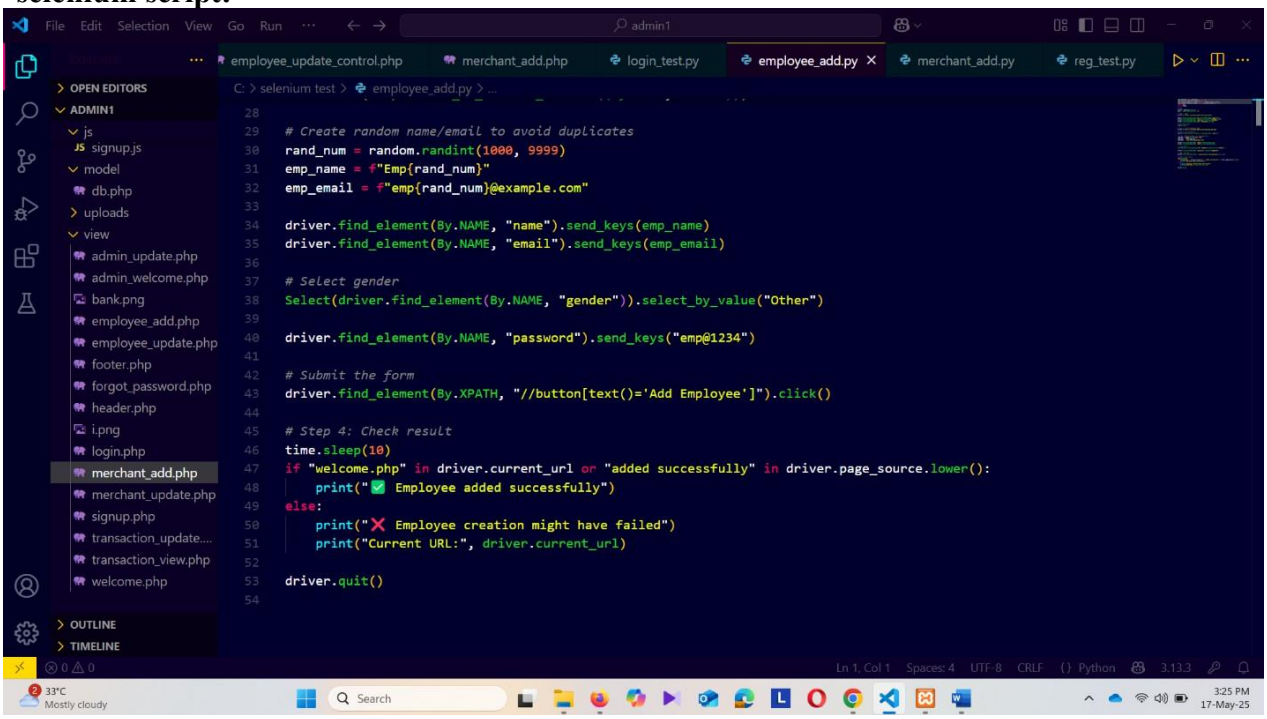
# Wait after login
time.sleep(5)

# Step 2: Go to Add Employee Page
driver.get("http://localhost/admin1/view/employee_add.php")

# Step 3: Fill out the form
wait.until(EC.presence_of_element_located((By.NAME, "name")))

# Create random name/email to avoid duplicates
rand_num = random.randint(1000, 9999)
```

selenium script:



```python
# Create random name/email to avoid duplicates
rand_num = random.randint(1000, 9999)
emp_name = f"Emp{rand_num}"
emp_email = f"emp{rand_num}@example.com"

driver.find_element(By.NAME, "name").send_keys(emp_name)
driver.find_element(By.NAME, "email").send_keys(emp_email)

# Select gender
Select(driver.find_element(By.NAME, "gender")).select_by_value("Other")

driver.find_element(By.NAME, "password").send_keys("emp@1234")

# Submit the form
driver.find_element(By.XPATH, "//button[text()='Add Employee']").click()

# Step 4: Check result
time.sleep(10)
if "welcome.php" in driver.current_url or "added successfully" in driver.page_source.lower():
    print("✅ Employee added successfully")
else:
    print("❌ Employee creation might have failed")
    print("Current URL:", driver.current_url)

driver.quit()
```

Fig_15: Employee Add – Selenium Script Output

Web view:

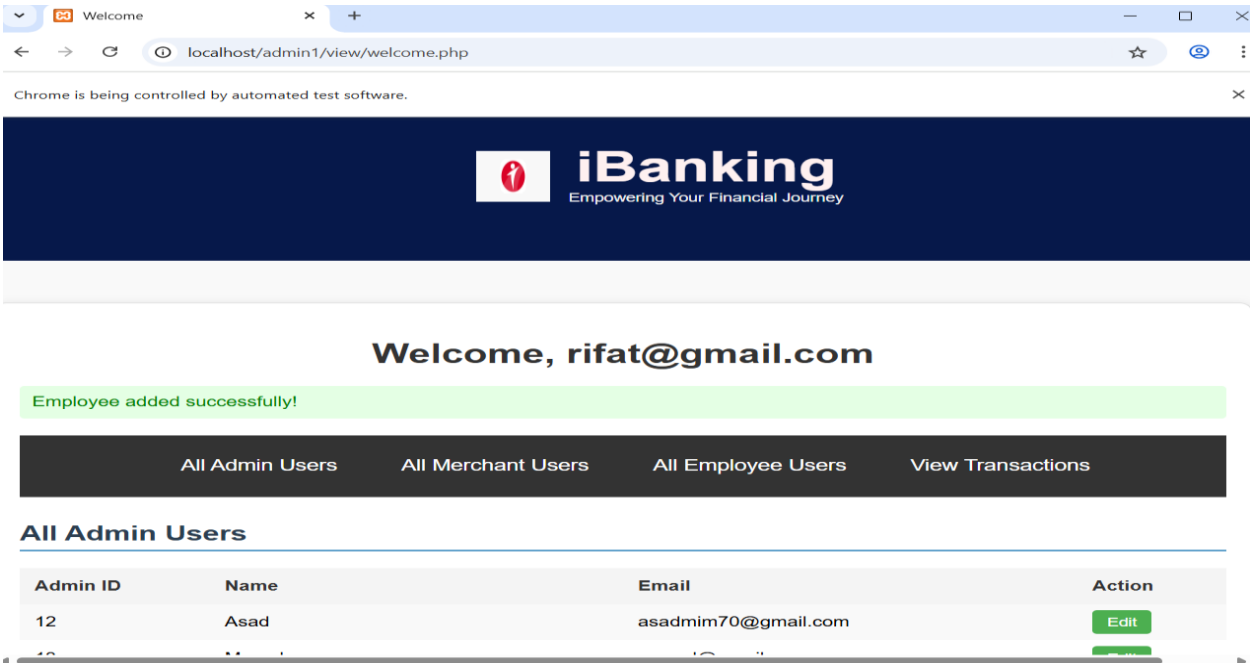
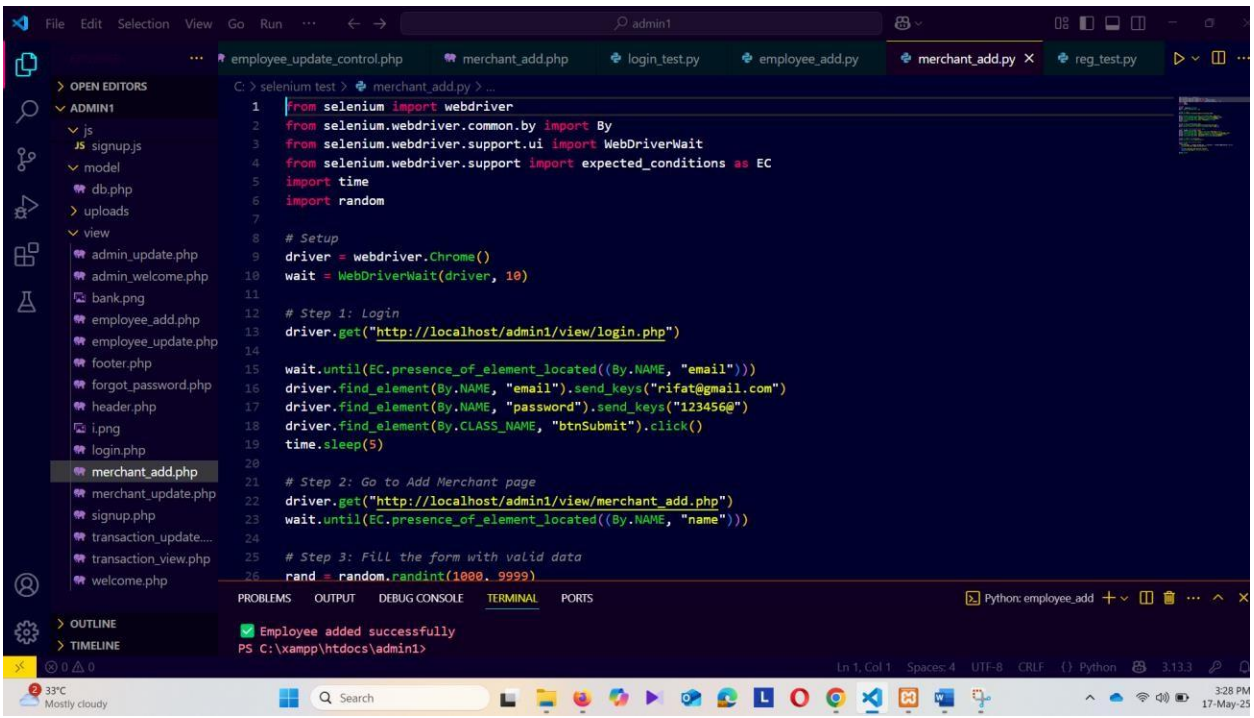


Fig_16: Employee Add – Web View

Case 4: Merchant Add:

selenium script:



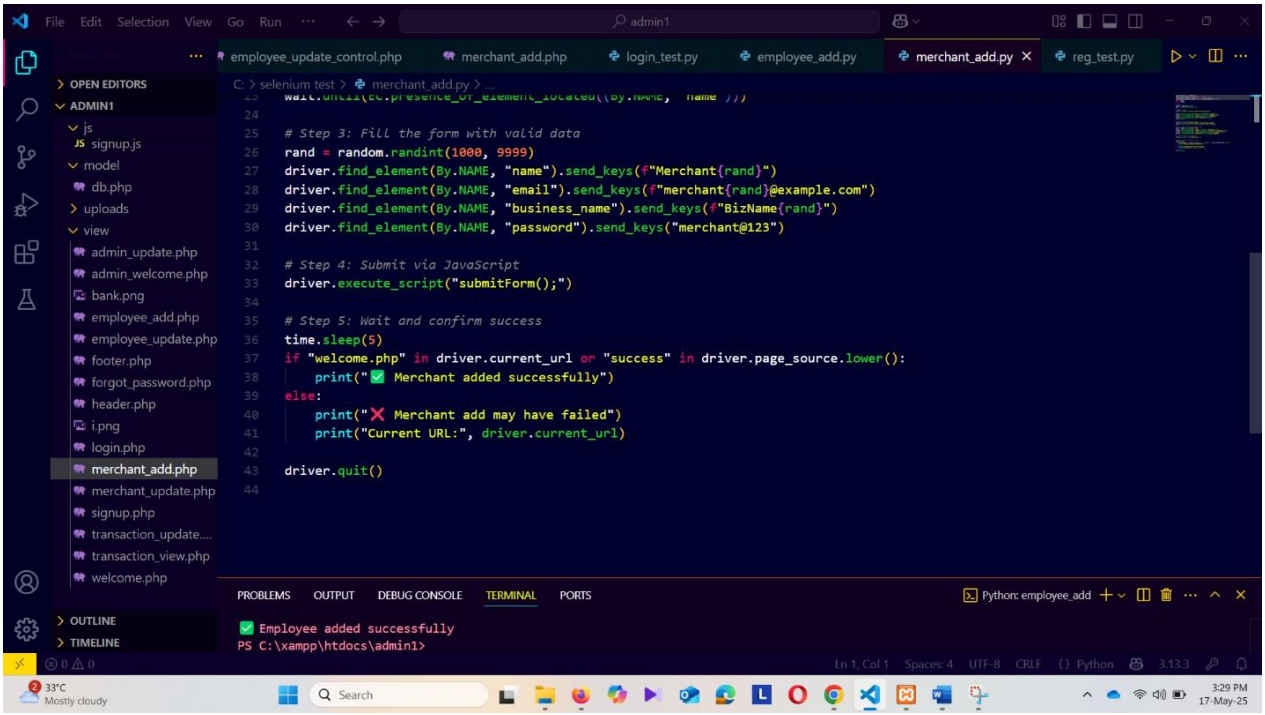


Fig_17: Merchant Add – Selenium Script Output

Web View:

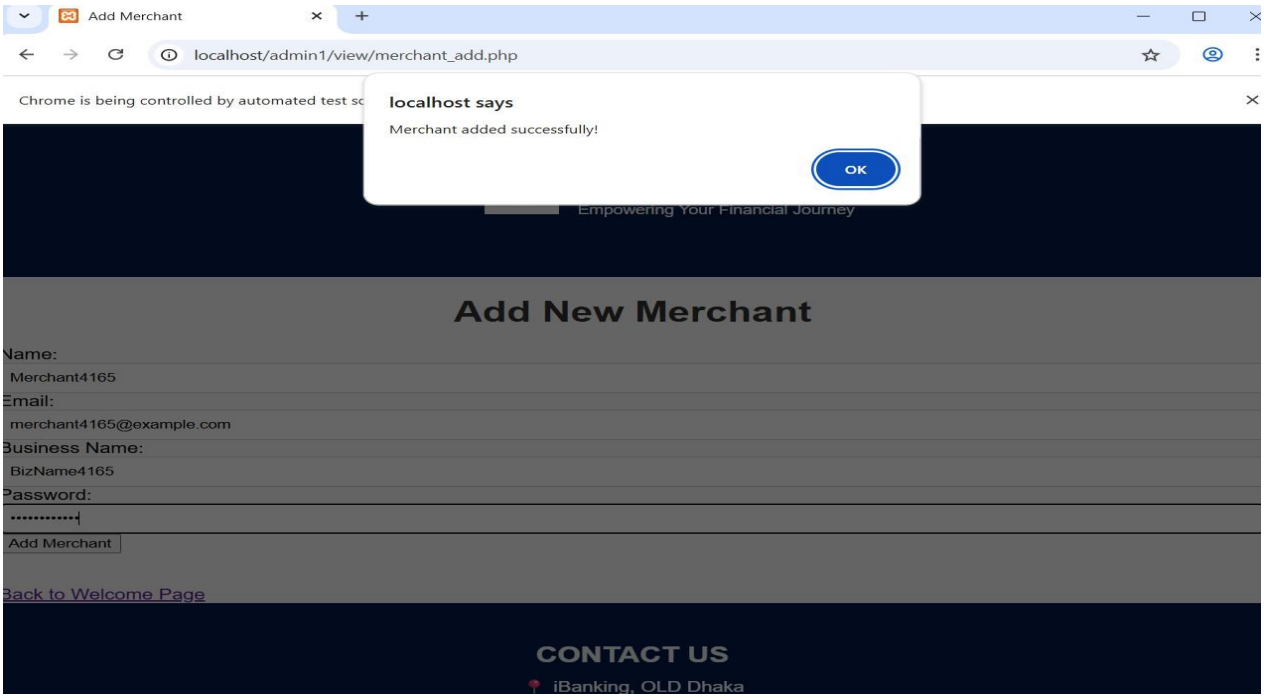


Fig_18: Merchant Add – Web View